MALAYSIA DATA INNOVATION TALENT x DOSM DATATHON 2025

"Data-Driven Insights on COVID-19 and Hospital Capacity: A Power BI Dashboard Approach"

by

DATA PHANTOMS

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1.0 INTRODUCTION

1.1 Background

Malaysian government hospitals are the cornerstone of the nation's healthcare system, coming under the Ministry of Health (MOH) and delivering most of the inpatient care to the nation. The hospitals are widely accessible and heavily subsidised but have long been plagued by problems of overcrowding, resource limitations, and rising demand as a result of an ageing population and chronic diseases (Ashraf, 2021; Hamzah et al., 2021).

During the COVID-19 pandemic, the MOH implemented measures to expand hospital capacity by assigning specific hospitals to treat COVID-19, redesignating existing wards, and centralising bed use reporting. The reforms were necessary since infection-control measures reduced the normal availability of beds, and demand soared. Reports show hospital services were reorganised, and resources were reassigned to areas of priority to deal with the influx of cases (Shah et al., 2020; Ang et al., 2021; MOH Annual Report, 2020).

Official records indicated that government hospital COVID-19 bed capacity reached a peak of over 6,000 during the peak of crises, with a few hospitals consolidating specialties and redirecting non-severe cases to quarantine centers to ease pressure. ICU and general ward quotas were also redistributed based on trends in cases. These efforts, together with limited private sector involvement, sustained hospital operations despite extreme pressure. However, analyses emphasize that the pandemic highlighted current capacity gaps and the urgent need for public hospital infrastructure and investment in critical care facilities (Amaran et al., 2021; Khor et al., 2020; Ang et al., 2021; Tan et al., 2021).

1.2 Problem Statement

The COVID-19 pandemic uncovered glaring weaknesses in the public hospitals' ability to handle bed space during a health emergency. Acute overcrowding was precipitated by a sudden surge in cases and their need for general wards and ICU beds outstripped existing capacity. In spite of measures like ward conversions and field hospital deployments on a temporary basis, many patients continued to experience delays in admission because of bed capacity. This identified a root systemic issue in planning for and managing bed shortages in the event of prolonged crises. The gradual release of non-availability of beds not only affected the care of patients but also placed undue pressures on medical personnel. All these issues call for change on resource re-allocator, upgrading of hospital facilities, and improved emergency readiness in Malaysia's health system.

1.3 Objectives

i. To prioritise data driven management systems at a larger scale for current and future development.

ii. To alleviate the issue of inconsistency of bed occupancy for patients in hospitals.

iii. To create a more convenient dashboard to monitor patient management in hospitals.

iv. To increase transparency of data to be accessible to stakeholders such as the hospital management and the government.

v. To strengthen hospital preparedness during emergencies or pandemics by identifying surge capacity and bed shortages early.

vi. To support the United Nations Sustainable Development Goals (SDGs), particularly SDG 3 (Good Health and Well-Being), by leveraging data analytics to enhance healthcare resilience and ensure equitable access to medical resources.

2.0 Literature Review

The COVID-19 pandemic posed the most significant communicable disease challenge to Malaysia since the 1918 Spanish Flu, which had a devastating mortality rate at that time. The COVID-19 outbreak in Malaysia commenced with the first confirmed case arriving on January 25, 2020, imported from China, marking the onset of the epidemic in the country (Hashim et al., 2021).

Epidemic Progression

Malaysia experienced multiple waves of COVID-19 infections. The first wave (January to mid-February 2020) was limited with 22 cases, primarily imported, and no deaths reported. The second wave began at the end of February 2020, primarily driven by clusters linked to mass gatherings, notably the Sri Petaling Tabligh cluster, the largest superspreading event that accounted for nearly half the nation's cases at one point. This cluster complicated containment and triggered local transmissions nationwide. The third wave, beginning October 2020, saw a surge in cases particularly in states like Sabah and Selangor, heavily taxing healthcare resources (Hashim et al., 2021).

Challenges Faced

Several challenges confronted Malaysia's healthcare system and public health response:

- Cluster Containment: The Sri Petaling cluster presented logistical challenges in contact tracing and mass testing due to incomplete attendee records and presence of undocumented migrants (Hashim et al., 2021).
- Healthcare Capacity: Concerns arose over hospital and ICU bed availability, PPE supply shortages, and healthcare workforce burnout. Malaysia responded with contingency hospital setups, procurement of ventilators, and PPE stockpiling (Hashim et al., 2021).

- Testing and Tracing: Early aggressive testing strategies targeted symptomatic and asymptomatic contacts alike, across diverse populations including refugees and migrant workers. Expansion to private labs alleviated testing burdens (Hashim et al., 2021).
- Community Transmission: Despite healthcare workers being high-risk, most infections among them were traced to community transmission rather than hospital exposures (Hashim et al., 2021).

Response Measures

- Movement Control Order (MCO): Instituted beginning March 18, 2020, the MCO was a phased nationwide lockdown that severely restricted movement, gatherings, and business operations. This intervention successfully flattened the epidemic curve, reducing peak active cases significantly below early projections (Hashim et al., 2021).
- Quarantine and Isolation: Designated facilities nationwide accommodated close contacts and travelers for mandatory quarantine, facilitating controlled monitoring and reducing home quarantine non-compliance (Hashim et al., 2021).
- Technology Utilization: MySejahtera and MyTrace mobile apps were deployed to support contact tracing, case detection, vaccine appointment scheduling, and public health communication (Hashim et al., 2021).
- Healthcare Worker Support: Mobilization of additional staff, inclusion of private and retired medical professionals, and mental health support initiatives helped manage healthcare workforce sustainability (Hashim et al., 2021).
- Vaccination Strategy: Malaysia committed to global vaccine access initiatives (such as COVAX) while also securing bilateral procurement agreements. The national vaccination plan prioritized healthcare workers, vulnerable populations, and adults systematically, aiming for herd immunity by early 2022 (Hashim et al., 2021).

Epidemiological Insights

The predominant SARS-CoV-2 strain in Malaysia was identified as strain B, common in East Asia, with other strains present as well. The virus exhibited mutations contributing to variable transmissibility. Vulnerable population groups, particularly adults aged 55 and above with comorbidities, faced higher morbidity and mortality risks (Hashim et al., 2021).

Impact and Lessons

The economic impact was considerable, with Malaysia losing estimated billions during the lockdown phase while balancing public health priorities. The government's multi-pronged and adaptive approach demonstrated resilience and coordination, offering valuable lessons in early preparedness, agile response, and the importance of community cooperation (Hashim et al., 2021).

3.0 Methodology

This study adopts a structured data analytics methodology to analyze the COVID-19 case data and hospital utilization records in Malaysia. The approach consists of five main stages: data collection, preprocessing, integration, exploratory data analysis, and analytical modeling/visualization.

3.1 Data Collection

Two primary datasets were utilized:

1. COVID-19 Cases Dataset – containing daily state-level records of new cases, recoveries, active cases, imported cases, and cluster-related cases.
2. Hospital Dataset – containing hospital-level statistics, including total beds, COVID-designated beds, admissions, discharges, and current hospitalizations.

Both datasets span the period from early 2020 onwards and provide state-level granularity.

3.2 Data Preprocessing

Prior to analysis, several preprocessing steps were undertaken:

- Data type conversion: The date fields were standardized into datetime format to allow temporal analysis.
- Handling missing values: Missing or inconsistent entries were inspected; where appropriate, forward-filling and zero imputation were applied to preserve continuity in the time series.

- Normalization: For comparative analysis across states, key variables (e.g., new cases, admissions) were normalized against population size and available hospital beds.
- Outlier detection: Unusual spikes in reported cases were identified and flagged for sensitivity analysis.

3.3 Data Integration

To facilitate joint analysis, the two datasets were merged on date and state, ensuring alignment between daily COVID-19 case counts and corresponding hospital capacity metrics. This integration allowed examination of the relationship between infection trends and hospital strain.

3.4 Exploratory Data Analysis (EDA)

EDA was conducted to identify key patterns and relationships:

- Capacity utilization: Calculation of hospital bed occupancy rates (% of total beds allocated and occupied by COVID patients).
- Correlation analysis: Assessing the relationship between case surges (cases_new) and subsequent hospital admissions (admitted_covid).
- State-level comparisons: Comparing metrics across states to identify regions most affected by hospital strain.

3.5 Analytical Modeling and Visualization

The analysis employed descriptive and inferential statistical methods supported by data visualization:

- Dashboards were developed to visualize trends in cases and hospital utilization dynamically.

4.0 Findings

4.1 Results

Our analysis of Malaysia's COVID-19 datasets (cases and hospital capacity from 2020–2025) revealed several key patterns:

1. Case Dynamics

    - Daily new cases were highly volatile, with extreme peaks exceeding 33,000 cases per day during major outbreak waves.
    - Active cases peaked above 323,000 nationally, putting severe strain on the healthcare system.
    - Imported cases were relatively low compared to local transmission, showing the outbreaks were largely community-driven.

2. Hospital Burden and Bed Utilization

    - Average national hospital bed utilization for COVID cases was 29.35%, indicating that while the system was resilient overall, stress was unevenly distributed.
    - Kuala Lumpur (49%), Penang (43%), and Selangor (40%) experienced critical utilization rates, highlighting disproportionate strain in urban centers.
    - Bed availability dynamically scaled during peak waves, but states with denser populations consistently experienced higher utilization pressures.

3. Admission and Discharge Trends

    - Hospital admissions and discharges tracked closely, suggesting capacity management strategies were in place to prevent overflow.
    - However, lag effects were observable: surges in new cases typically preceded increased hospitalizations by 7–14 days.
    - This lag provides a predictive opportunity for proactive resource allocation.

4. Correlation Between Cases and Hospital Strain

    - Statistical correlation showed strong positive relationships between:
        - cases_new and hosp_covid
        - cases_active and bed utilization %.
    - This confirms that infection waves directly translate into hospitalization surges, validating the dashboard KPI design.

Information Architecture / Solution Approach

To transform raw COVID-19 and hospital bed datasets into actionable insights, we designed a multi-layered analytics pipeline:

1. Data Integration Layer
    - Combined two datasets (covid_cases, hospital_beds) using date and state as primary keys.
    - Performed data cleaning and normalization to correct anomalies (e.g., negative active case counts due to backdated adjustments).
    - Created derived features such as bed utilization %, 7-day moving averages, and lag variables to capture predictive dynamics.

2. Analytical Layer
    - Descriptive analytics: Identified historical peaks, recovery waves, and geographic hotspots.
    - Diagnostic analytics: Correlated new cases with hospitalization lag patterns.
    - Predictive analytics: Time-series modeling (ARIMA/LSTM) applied to forecast hospital strain based on new case trajectories.

3. Visualization & Dashboard Layer
    - Built an interactive Power BI dashboard integrating:
        - National & state-level active cases.
        - Hospital bed utilisation vs. capacity.
        - Admission vs. discharge timelines.
    - Designed intuitive KPI cards (e.g., total active cases, utilisation rate) alongside line/area charts to highlight temporal trends.

4. Solution Architecture for Decision-Making
    - Early Warning System: By monitoring cases_new and applying a predictive lag, health authorities can anticipate bed shortages up to 2 weeks in advance.
    - Resource Allocation Engine: The architecture supports real-time decision-making by flagging states nearing critical utilisation thresholds (>40%).
    - Scalability: The modular pipeline allows integration of vaccination, ICU, or demographic datasets to refine forecasts.

5.0 CONCLUSION

The COVID-19 crisis revealed both the indispensable role of Malaysian government hospitals and the structural weaknesses that limit their resilience. As the backbone of the Ministry of Health's public healthcare system, these hospitals provide most inpatient services at low cost but already faced overcrowding, staff shortages, and rising demand from chronic illness and ageing populations (Ashraf, 2021; Hamzah et al., 2021). The pandemic pushed ward and intensive care unit (ICU) capacity beyond limits, forcing rapid reforms. The Ministry designated COVID-19 hospitals, converted wards, and centralised bed tracking to meet demand (Shah et al., 2020; Ang et al., 2021). While these measures boosted surge capacity, they also reduced non-COVID services, exposing systemic fragility (MOH, 2020). At the peak, over 6,000 public hospital beds were allocated to COVID-19 patients, with mild cases redirected to quarantine centres (Amaran et al., 2021; Khor et al., 2020). These strategies prevented collapse but showed dependence on reactive rather than sustainable solutions. Lessons highlight the need for stronger baseline capacity, ICU preparedness, and resource distribution. Scholars also emphasise combining digital innovations with infrastructure (Ang et al., 2021; Tan et al., 2021). In this regard, our hospital bed management dashboard integrates real-time data on admissions, discharges, and ICU use to improve transparency and decision-making. Predictive analytics forecast demand, enabling proactive planning, better coordination, reduced transfer delays, and balance between COVID-19 and routine services. Beyond crises, the tool supports daily hospital management, optimising capacity and equitable care. Malaysia's experience shows resilience depends on pairing infrastructure with digital solutions to build efficiency, responsiveness, and preparedness for future emergencies.

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Consultation session between our advisor and Dr Mohd Izzun Nasheef bin Mohamad Hasmuri

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